Hotel Booking Cancellation Analysis Report

Introduction

Hotel booking cancellations are a significant challenge in the hospitality industry. When customers cancel reservations, hotels may experience revenue loss, inefficient room allocation, and difficulties in demand forecasting. Understanding the reasons behind booking cancellations is essential for improving business strategies and operational efficiency.

This project analyzes hotel booking data to identify patterns and factors that influence cancellations. Through exploratory data analysis (EDA) and data visualization, the study aims to uncover insights that can help hotels reduce cancellation rates and improve overall booking management.

Objectives

The main objectives of this analysis are:

- To analyze hotel booking data and understand customer booking behavior.
- To determine the percentage of bookings that are canceled.
- To compare cancellation patterns between different hotel types.
- To examine how factors such as lead time, pricing, and booking channels affect cancellations.
- To identify trends and patterns that can help hotels reduce cancellations.

Data Cleaning and Preprocessing

Before performing the analysis, the dataset was cleaned and prepared.

Steps Performed

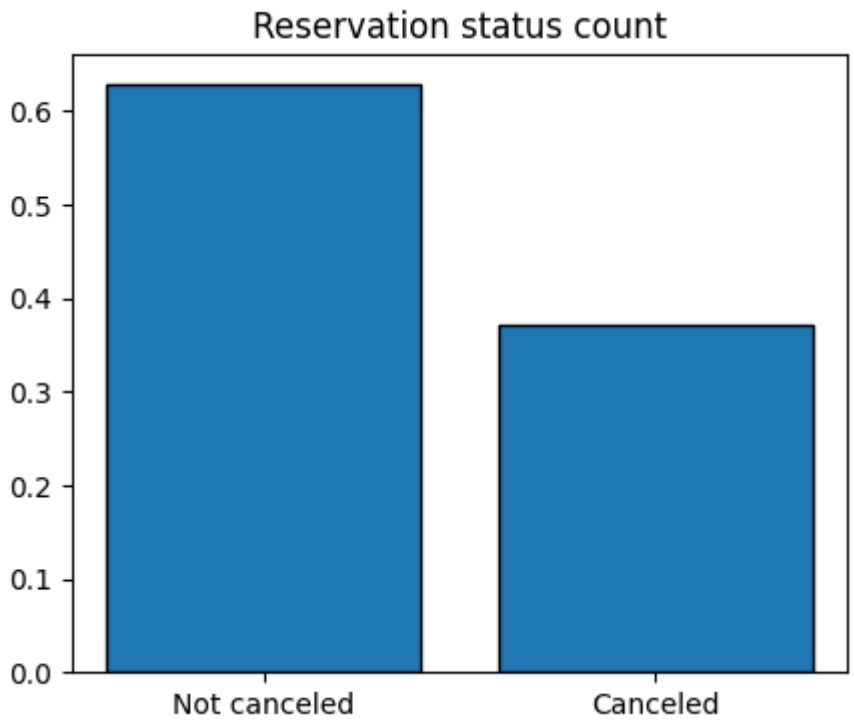
- Loaded dataset using Python and Pandas
- Checked for missing values
- Removed unnecessary columns
- Converted date-related columns into proper date formats
- Handled missing values where required
- Ensured data consistency

Exploratory Data Analysis (EDA)

Exploratory Data Analysis was conducted to understand the dataset and identify patterns related to hotel bookings and cancellations.

Booking Cancellation Distribution

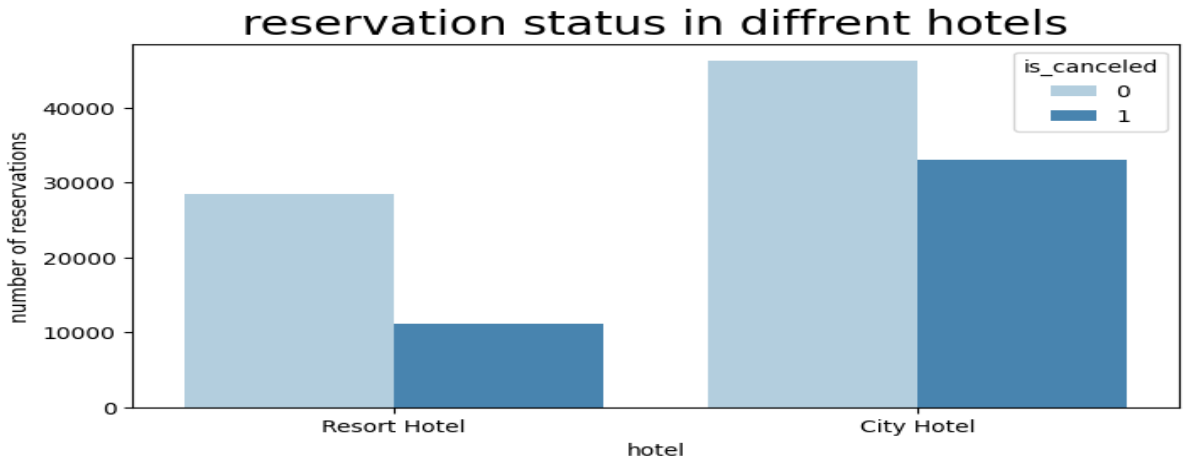
This analysis examines the number of canceled bookings compared to non-canceled bookings.



Explanation:
The visualization shows the proportion of canceled and non-canceled bookings in the dataset.

Cancellation by Hotel Type

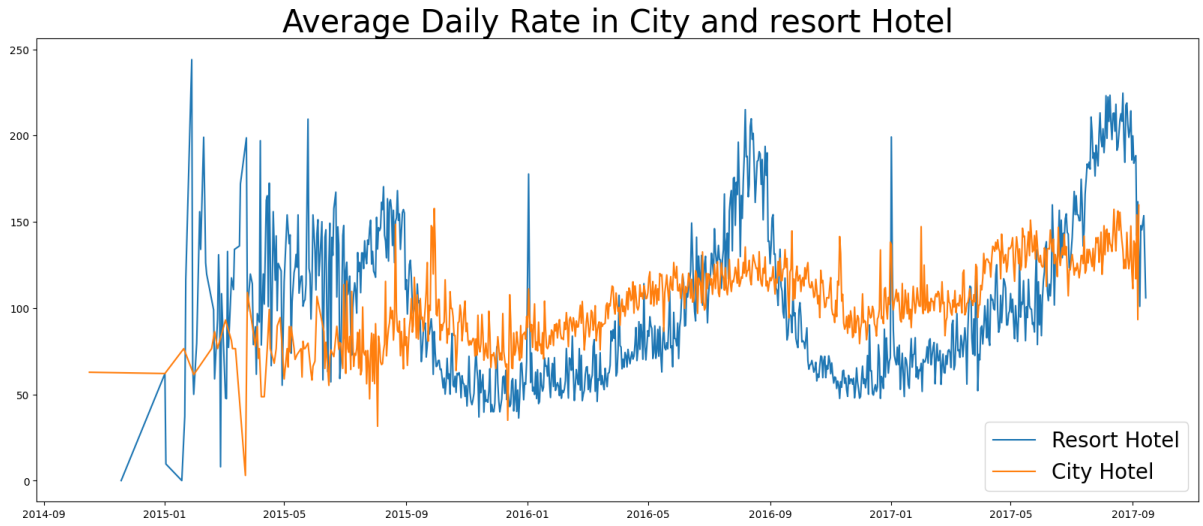
This analysis compares the cancellation rates between different hotel types.



Explanation:
The chart highlights which hotel type experiences higher cancellation rates.

Average Daily Rate in City Hotel and Resort Hotel

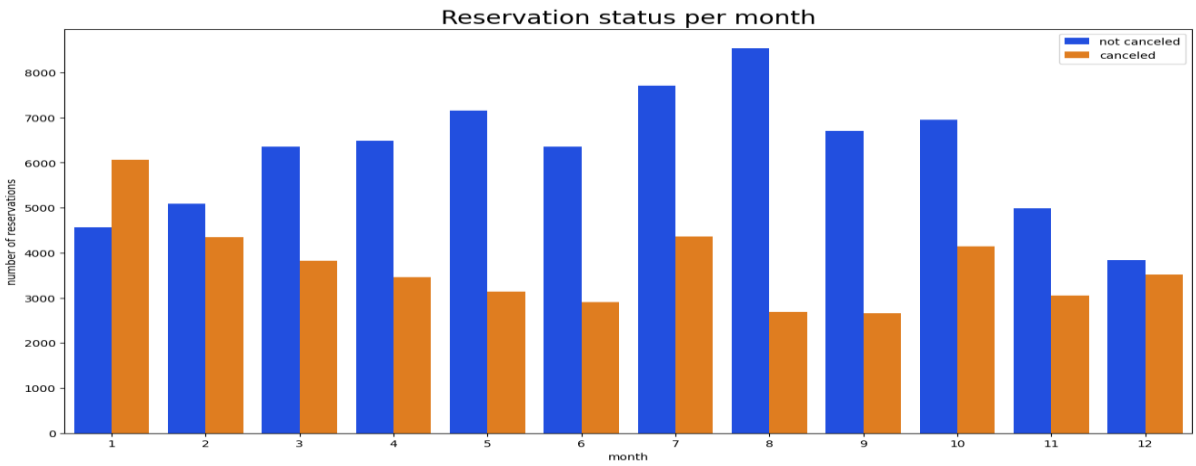
This visualization compares the **Average Daily Rate (ADR)** between City Hotels and Resort Hotels over time. ADR represents the average revenue earned per occupied room and helps understand pricing patterns between different hotel types.



Explanation:
The line chart shows the ADR trends for both hotel types. It helps identify differences in pricing strategies and seasonal fluctuations between City Hotels and Resort Hotels.

Reservation Status per Month

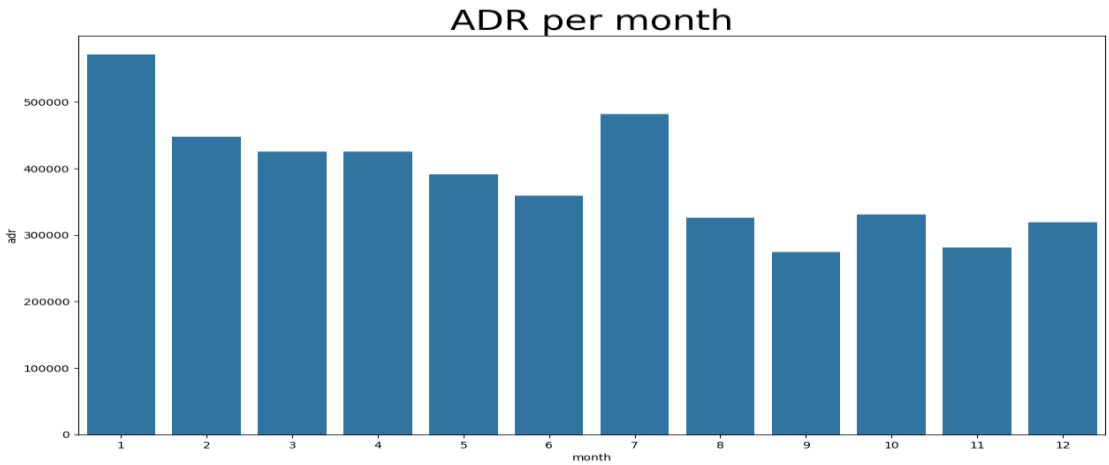
This visualization shows the **number of reservations made each month**, separated by whether the booking was canceled or not canceled.



Explanation:
The chart helps identify seasonal booking trends and understand which months experience higher cancellation rates. This insight is useful for hotels to adjust pricing and promotional strategies during different times of the year.

Average Daily Rate (ADR) per Month for Canceled Bookings

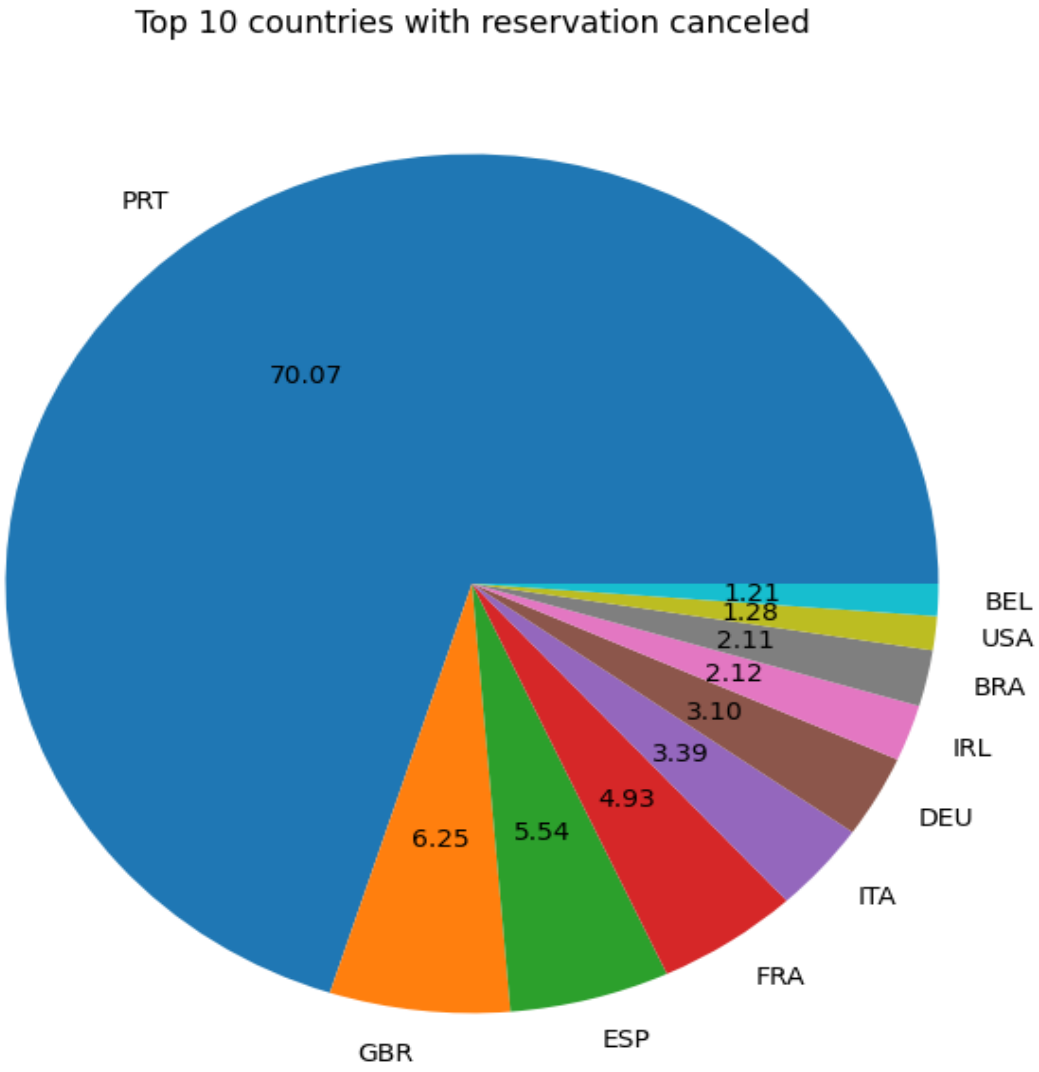
This visualization examines the **monthly ADR values for bookings that were canceled**.



Explanation:
The bar chart highlights the total ADR values associated with canceled bookings across different months. This helps determine whether higher room prices influence cancellation behavior.

Top 10 Countries with Reservation Cancellations

This visualization shows the **top 10 countries where customers canceled hotel reservations**.



Explanation:
The pie chart represents the proportion of cancellations from different countries. This helps identify geographical regions with higher cancellation rates and can assist hotels in targeting marketing or policy adjustments accordingly.

Key Insights

The analysis revealed several important insights:

- City hotels experience higher cancellation rates compared to resort hotels.
- Customers who book far in advance are more likely to cancel their reservations.

- Bookings made through online travel agencies contribute significantly to cancellations.
- Seasonal demand patterns influence hotel booking trends.
- Customers who make bookings without deposits tend to cancel more frequently.

Business Recommendations

Based on the findings, the following recommendations can help hotels reduce cancellations:

- Implement deposit policies to discourage last-minute cancellations.
- Use dynamic pricing strategies during peak seasons.
- Encourage direct bookings to reduce reliance on third-party platforms.
- Offer discounts or incentives for non-refundable bookings.
- Improve communication with customers before arrival to reduce cancellation risk.